

SEAN KANE

Syosset, NY

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Education

Stony Brook University

August 2023 – May 2028

Bachelor of Engineering in Electrical Engineering (GPA: 3.8 / 4.0); Expected May 2027

Stony Brook, NY

Master of Science in Electrical Engineering; Expected May 2028

Coursework: Advanced VLSI Systems Design, Advanced Digital System Design & Generation

Technical Skills

HDLs & Programming: SystemVerilog, VHDL, SystemRDL, Python, C, C++, Embedded C

EDA & Design Tools: Siemens QuestaSim, Synopsys Design Compiler, Xilinx Vivado, Cadence Allegro, Altium Designer

Verification & Tools: Linux, Git, UVM RAL, IP-XACT, JTAG/SWD, Logic Analyzer, Oscilloscope

Experience

FPGA Engineering Intern

June 2026 – August 2026

Collins Aerospace

Cedar Rapids, IA

- Built a Python CLI pipeline that converts SystemRDL into IP-XACT, VHDL register RTL, C headers, HTML documentation, and UVM RAL, reducing a multistep FPGA/ASIC register-generation workflow to one command.
- Customized IP-XACT exports with vendor-specific metadata for hardware-access behavior and functional-coverage configuration, preserving register semantics across downstream design and verification tools.
- Packaged and validated the toolchain for offline Windows and Linux deployment using local Python dependencies, shell launchers, structured logging, and automated execution checks.

Operations Engineer Intern

May 2025 – August 2025

TC Electric, LLC

New York, NY

- Designed a Power BI system that ingested field-captured equipment and sensor data, automating asset tracking and usage statistics across projects and reducing operational errors and costs by \$2,000.
- Developed a Python automation workflow to migrate aerial survey data from a restricted, pay-per-user platform to OpenSpace Air, expanding company-wide access to project data.

President

May 2025 – May 2026

Institute of Electrical and Electronics Engineers (IEEE) Stony Brook University Branch

Stony Brook, NY

- Organized 20+ technical workshops and engineering outreach events per semester, including the 2025 IEEE Regional Student Conference for 250+ attendees focused on professional development and applied engineering skills.
- Designed and led a hands-on embedded systems workshop covering I2C, SPI, and UART, guiding students through microcontroller-to-sensor interfacing, register-level programming, and logic-analyzer-based debugging.

Projects

2D Convolution Accelerator | *SystemVerilog, RTL Design, RTL Verification*

August 2026 – Present

- Designing a synthesizable SystemVerilog hardware accelerator to perform 2D convolution, translating the computational algorithm into register-transfer-level digital logic.
- Developing and verifying modular RTL components using SystemVerilog testbenches and randomized simulation in Siemens QuestaSim.
- Synthesizing the accelerator with Synopsys Design Compiler targeting the Nangate 45 nm standard-cell library to evaluate the hardware implementation and synthesis results.

HP 5004A Signature Analyzer | *VHDL, FPGA Design, Digital Logic*

May 2026

- Designed a VHDL-based HP5004A digital signature analyzer using a 16-bit LFSR to compact serial input data into repeatable hexadecimal signatures for hardware fault-analysis workflows.
- Implemented modular RTL components including start/stop gate-generation FSMs, edge-select sampling logic, LFSR control, output buffering, and multiplexed 7-segment display driving.
- Developed VHDL simulation testbenches and validated the implemented design on a Xilinx FPGA, verifying timing, state transitions, signature capture, and real-time display output.

Dual-Mode BLE Beacon | *RF PCB Design, Altium Designer, 2.4 GHz MIFA*

December 2025

- Engineered a compact 1.7" × 1.0" 4-layer nRF52832 BLE beacon PCB with a 2.4 GHz MIFA antenna, applying RF layout constraints including antenna keepout, no-copper regions, and via fencing in a CR2032-powered form factor.
- Developed embedded C firmware with the Nordic SDK implementing dual Finder/Game modes and a BLE advertising state machine, and programmed/debugged the board over SWD/JTAG using a Raspberry Pi Pico as a probe.